

Practical - 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem definition: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

Theory

1. How are DOM elements selected and modified?

Answer: DOM elements are selected using `getElementById()` and `querySelectorAll()` and then their content or style is changed using JavaScript.

2. Are event listeners attached correctly?

Answers: Yes, event listeners are attached using `addEventListener()` to handle clicks on buttons and other elements.

3. Is local Storage used for remembering theme choice?

Answers: Yes, local Storage is used to save the selected light or dark theme and keep it after refreshing the page.

4. Does interactivity improve usability without breaking accessibility?

Answers: Yes, the interactive features make the website easier to use, while buttons, labels and alt text help maintain accessibility.

A. Implementation/ Output/Result (Keep only one option which is suitable to practical)

A. HTML tags used are:

Tag	Application / Use
<code><!DOCTYPE html></code>	Tells the browser that the document uses HTML5.
<code><html></code>	Root element of the HTML page.
<code><head></code>	Contains page information and links to CSS.
<code><meta></code>	Provides information like character encoding and responsive viewport.
<code><title></code>	Sets the title shown on the browser tab.
<code><body></code>	Contains all visible content of the webpage.
<code><header></code>	Defines the top/header section of the webpage.

Tag	Application / Use
<div>	Used to group and organize elements.
	Displays an image such as the StudentHub logo or event images.
<h1>	Main heading of the page.
<h2>	Subheading or section heading.
<h3>	Smaller heading, used for event titles/features.
<nav>	Contains navigation links.
<a>	Creates links between pages.
<main>	Contains the main content of the webpage.
<section>	Groups related content into sections.
<article>	Represents an individual piece of content, such as a feature.
<p>	Displays paragraphs/text.
<button>	Creates clickable buttons for JavaScript interactions.
<footer>	Defines the bottom/footer section of the webpage.
<script>	Connects the external JavaScript file to HTML.

B. Attributes used are:

Attribute	Application
lang	Specifies the language of the webpage.
charset	Defines character encoding, usually UTF-8.
name	Used in the viewport meta tag.
content	Gives the value for the viewport setting.
id	Gives a unique name to an element so JavaScript can select it.
class	Groups elements for CSS styling or JavaScript selection.
src	Specifies the path of an image or JavaScript file.
alt	Provides alternative text for an image for accessibility.
href	Specifies the path of another webpage or CSS file.
aria-label	Gives an accessible name to buttons/elements.
role	Defines the purpose of an element for accessibility.
aria-modal	Tells assistive technology that an element is a modal dialog.
aria-labelledby	Connects a modal to its heading for accessibility.

C. CSS selectors used are

CSS	Application
*	Selects all elements.
body	Styles the complete webpage.
.class	Selects elements having a particular class.
#id	Selects an element with a particular ID.

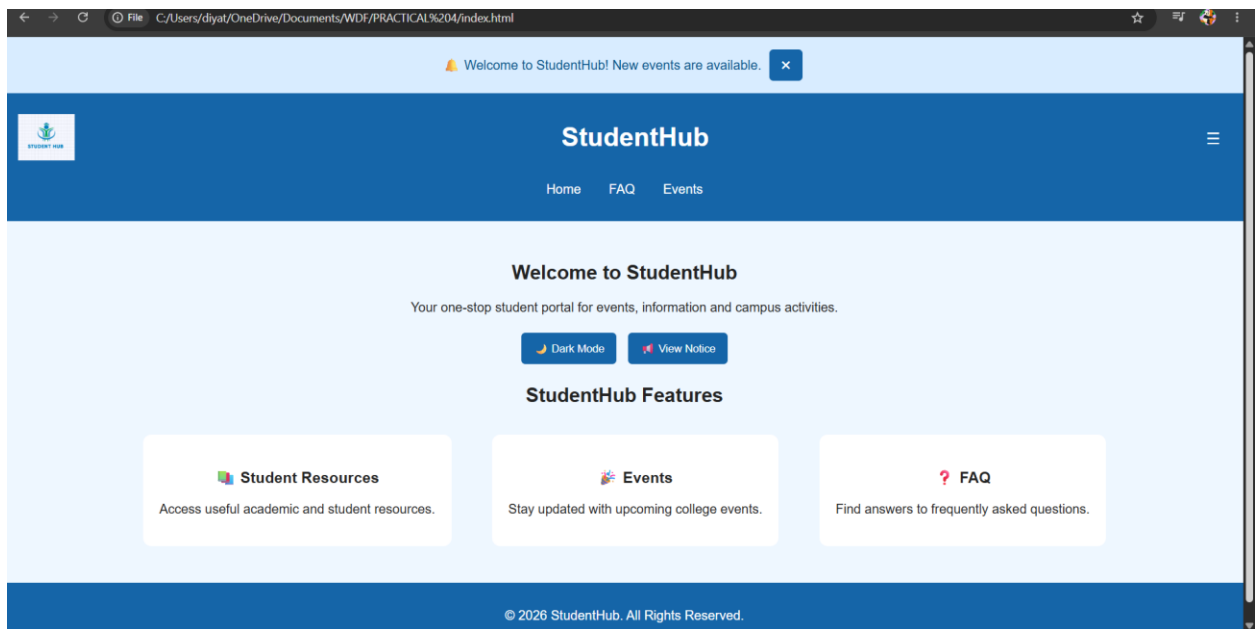
CSS	Application
background	Sets background color/image.
color	Sets text color.
padding	Adds space inside an element.
margin	Adds space outside an element.
font-family	Sets the font.
font-size	Changes text size.
display	Controls how an element is displayed.
width	Sets element width.
height	Sets element height.
border	Adds a border.
border-radius	Makes corners rounded.
cursor	Changes the mouse cursor.
text-align	Aligns text.
flex / display: flex	Arranges elements in a row/column.
gap	Adds space between flex items.
object-fit	Controls how an image fits inside its container.
position	Controls positioning of elements.
transition	Gives a smooth visual change.

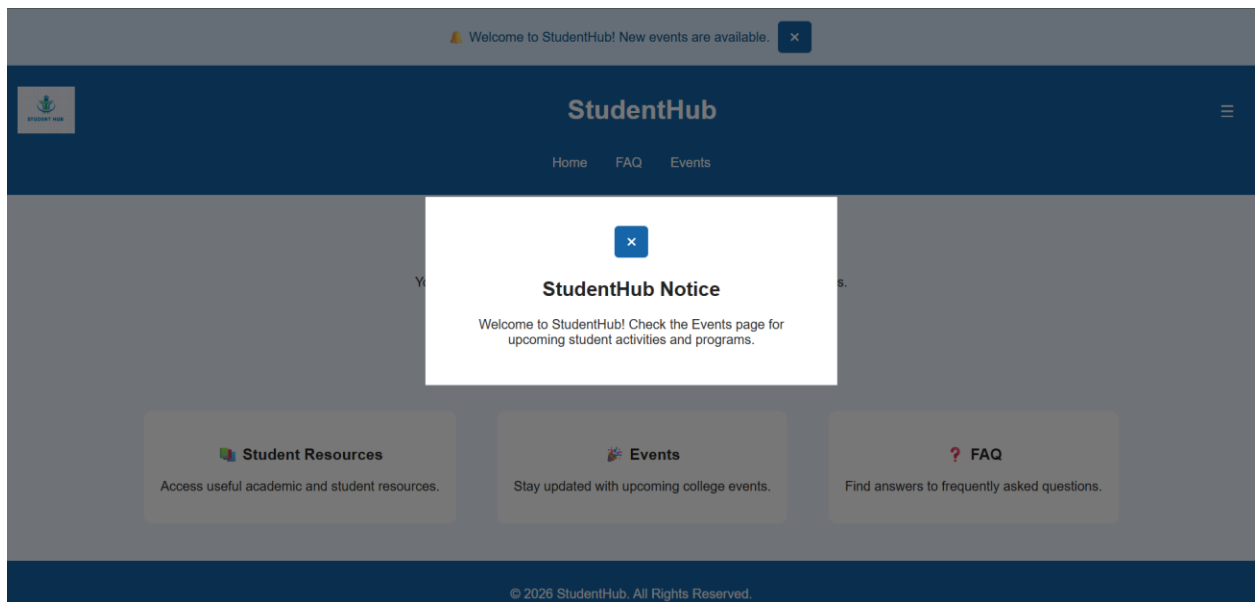
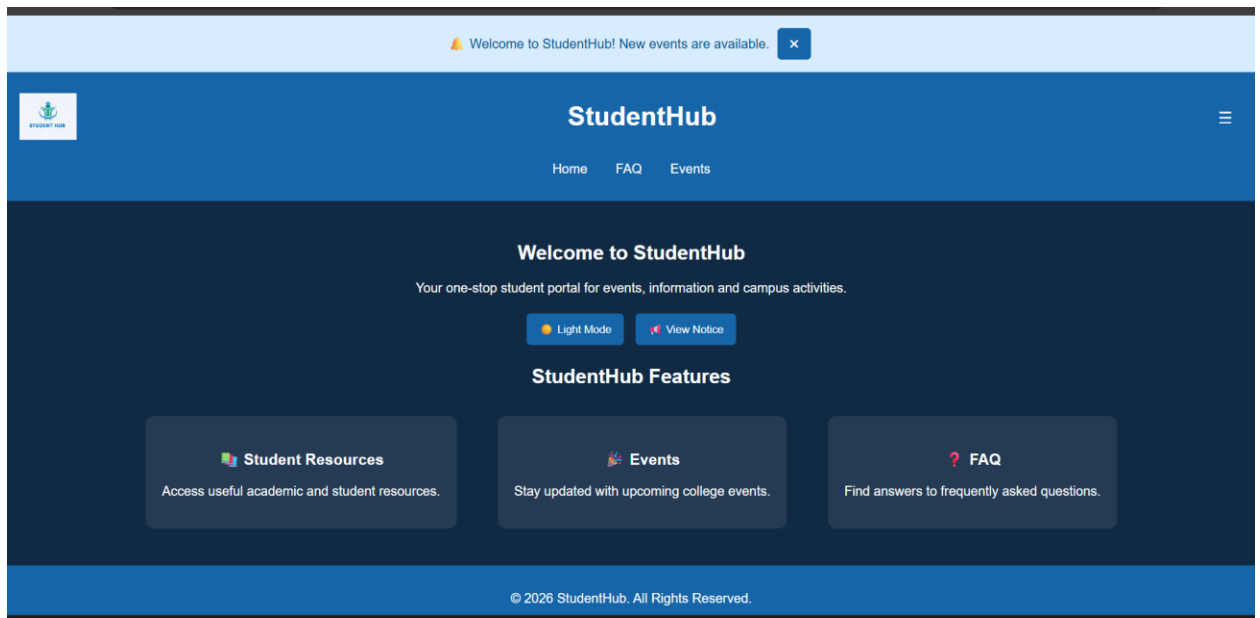
D. Javascript used are:

JavaScript	Application
let	Creates a variable.
document	Represents the HTML document.
getElementById()	Selects an element using its ID.
querySelectorAll()	Selects multiple elements using a CSS selector.
addEventListener()	Performs an action when an event occurs.
"click"	Event that occurs when an element is clicked.
classList	Accesses the classes of an element.
classList.toggle()	Adds/removes a class.
classList.contains()	Checks whether an element has a particular class.
style.display	Shows or hides an element.
innerHTML	Changes HTML content inside an element.
nextElementSibling	Gets the next HTML element.
localStorage	Stores data in the browser.
setItem()	Saves data in localStorage.
getItem()	Retrieves data from localStorage.
if	Checks a condition.
else	Executes code when the condition is false.

JavaScript	Application
forEach()	Performs an action for each selected element.
function	Creates a reusable block of code.
currentSlide	Stores the current slider position.
++	Increases a value by 1.
--	Decreases a value by 1.

A. Evidence of Work (Screenshots)





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Frequently Asked Questions

Click on a question to view its answer.

What is StudentHub?

How can I register on StudentHub?

How can I view upcoming events?

Is StudentHub available for all students?

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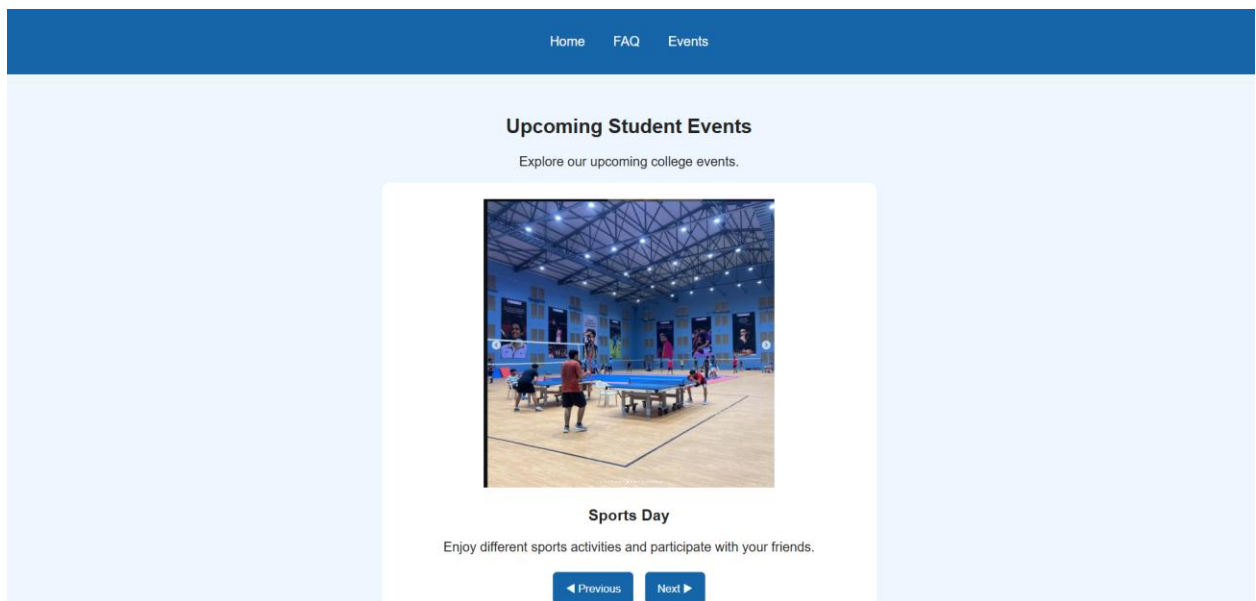
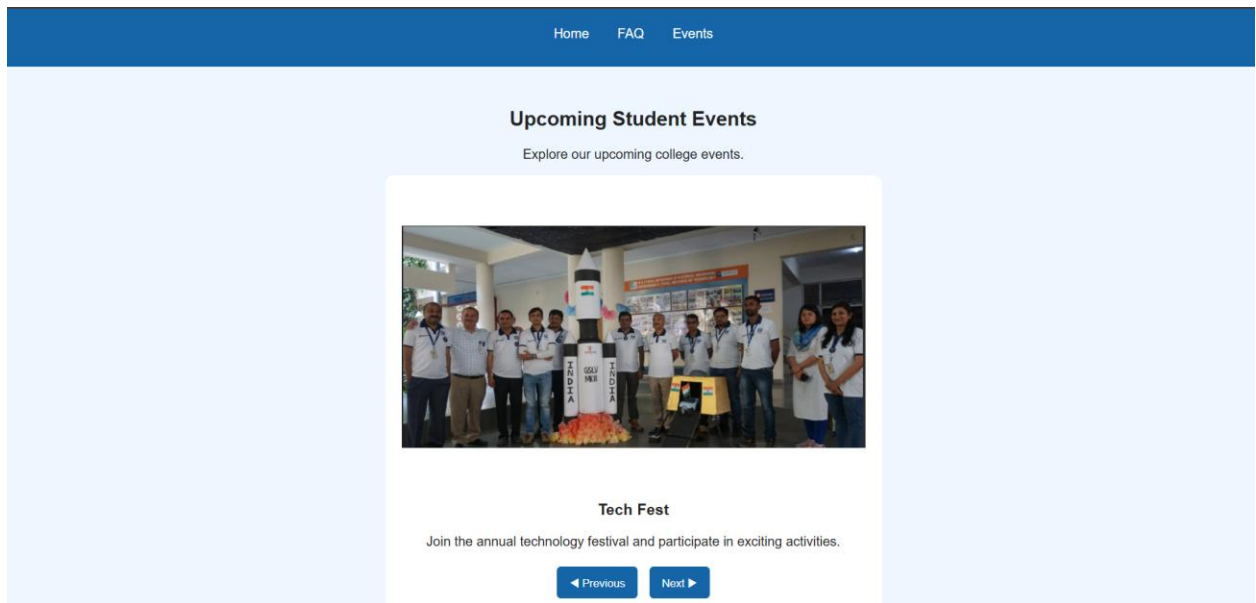
StudentHub is a student portal that provides useful information, resources and campus events.

How can I register on StudentHub?

How can I view upcoming events?

Is StudentHub available for all students?

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A. GitHub Repository Link:

<https://github.com/diyatolia1212/StudentHub/tree/b43e1228bd8cef4c5c558c2b97de3cfc2ee657f4/PRACTICAL%204>

B. Challenges Faced:

- Understanding how DOM elements are selected using JavaScript.
- Handling click events correctly.

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- Implementing the image slider and FAQ functionality.
- Managing light/dark mode using localStorage.

Learning Outcomes:

- Learned to use JavaScript for DOM manipulation and event handling.
- Learned to create interactive features like FAQ, modal, slider and hamburger menu.
- Learned to use localStorage for saving theme preferences. Improved understanding of basic UI interactivity and accessibility.

Faculty Remarks: (Keep this blank for faculty)

